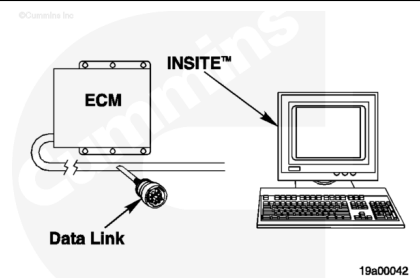


Trainings ▾

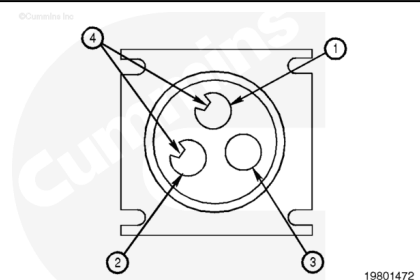
General Information

The RS232 datalink circuit is used by INSITE™ for CENSE™ to communicate with the CENSE™ ECM.



The datalink uses a 3-pin Deutsch connector. The wiring positions follow:

1. Pin A: - Ground
2. Pin C: - Datalink Transmit (Tx)
3. Pin B: - Datalink Receive (Rx)
4. Key



The procedures that follow cover the cab-located datalink connector and the engine-side datalink connector.

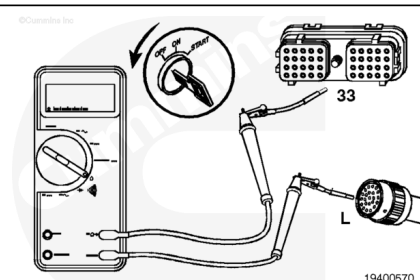
Resistance Check

Engine Located CENSE™

Disconnect the OEM harness from the CENSE™ 23-pin OEM connector. Remove the engine-located CENSE™ harness ECM A and B connectors from the ECM. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).

Use test leads, Part No. 3822758, on the ECM connector and Part No. 3824811, on the 23-pin Deutsch connector. Turn the keyswitch OFF.

Measure the resistance from pin 33 of the CENSE™ harness ECM A connector to pin L of the 23-pin Deutsch connector. The multimeter **must** show a closed circuit (10 ohms or less). If the



circuit is **not** closed, repair or replace the CENSE™ harness.

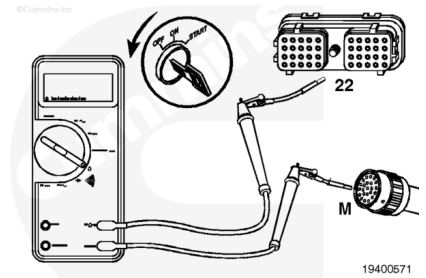
Refer to Procedure 019-043

(/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).

Measure the resistance from pin 22 of the CENSE™ harness ECM A connector to pin M of the 23-pin Deutsch connector. The multimeter **must** show a closed circuit (10 ohms or less).

If the circuit is **not** closed, repair or replace the CENSE™ harness. Refer to Procedure 019-043

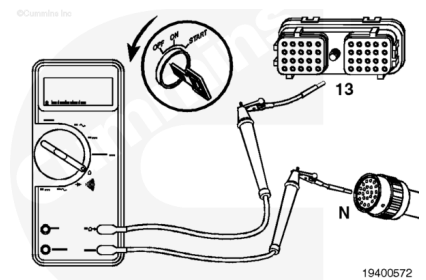
(/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).



Measure the resistance from pin 13 of the CENSE™ harness ECM B connector to pin N of the 23-pin Deutsch connector. The multimeter **must** show a closed circuit (10 ohms or less).

If the circuit is **not** closed, repair or replace the engine harness. Refer to Procedure 019-043

(/qs3/pubsys2/xml/en/procedures/81/81-019-043.html). If all measurements are within specifications, the OEM harness **must** still be checked. Refer to the OEM manual.



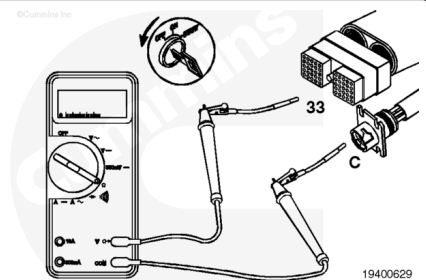
Disconnect the OEM harness from the CENSE™ 23-pin OEM connector. Remove the engine-located CENSE™ harness ECM A and B connectors from the ECM. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).

Use test leads, Part No. 3822758, on the ECM connector and, Part No. 3824811, on the 3-pin Deutsch connector. Turn the keyswitch OFF.

Measure the resistance from pin 33 of the CENSE™ harness ECM A connector to pin C of the 3-pin Deutsch connector. The multimeter **must** show a closed circuit (10 ohms or less).

If the circuit is **not** closed, repair or replace the CENSE™ harness. Refer to Procedure 019-043

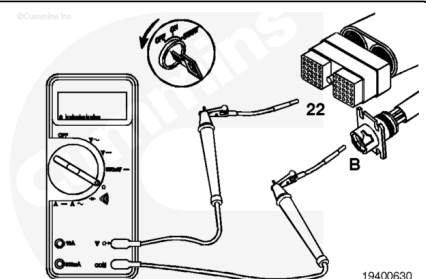
(/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).



Measure the resistance from pin 22 of the CENSE™ harness ECM A connector to pin B of the 3-pin Deutsch connector. The multimeter **must** show a closed circuit (10 ohms or less).

If the circuit is **not** closed, repair or replace the CENSE™ harness. Refer to Procedure 019-043

(/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).



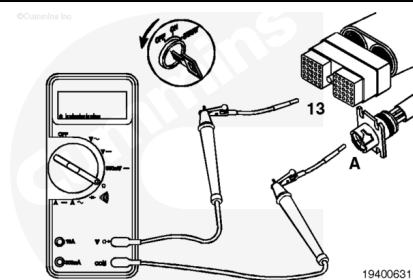
Measure the resistance from pin 13 of the CENSE™ harness ECM B connector to pin A of the 3-pin Deutsch connector. The multimeter **must** show a closed circuit (10 ohms or less).

If the circuit is **not** closed, repair or replace the engine harness.

Refer to Procedure 019-043

(/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).

If all measurements are within specifications, the OEM harness **must** still be checked. Refer to the OEM manual.



Check for Short Circuit to Ground

Cab-Located CENSE™

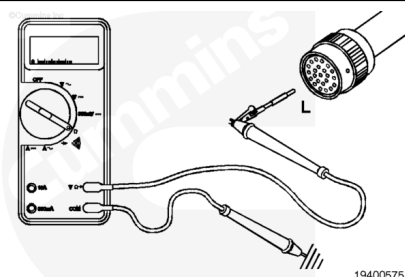
Use test lead, Part No. 3824811, for the cab located 23-pin Deutsch connector. Disconnect the 23-pin Deutsch OEM connector. Disconnect the ECM A and B connectors.

Measure the resistance from pin L of the 23-pin Deutsch connector to the engine block ground. The multimeter **must** show an open circuit (100k ohms or more).

If the circuit is **not** open, repair or replace the CENSE™ harness.

Refer to Procedure 019-043

(/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).

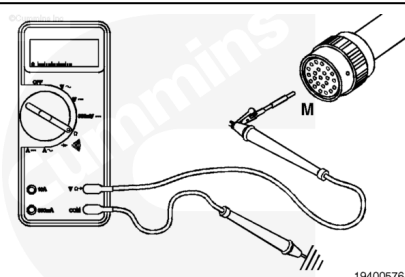


Measure the resistance from pin M of the 23-pin Deutsch connector to the engine block ground. The multimeter **must** show an open circuit (100k ohms or more).

If the circuit is **not** open, repair or replace the CENSE™ harness.

Refer to Procedure 019-043

(/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).

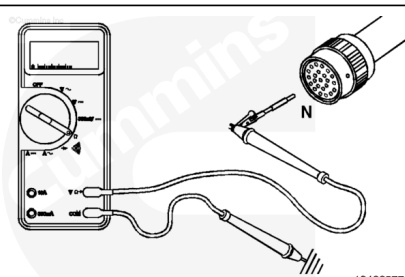


Measure the resistance from pin N of the 23-pin Deutsch connector to the engine block ground. The multimeter **must** show an open circuit (10 ohms or less).

If the circuit is **not** open, repair or replace the CENSE™ harness.

Refer to Procedure 019-043

(/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).

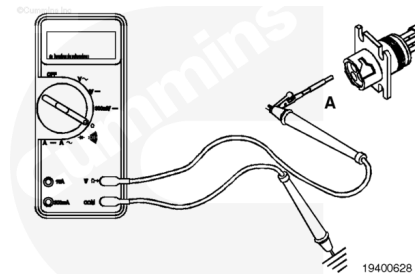


Engine-Located CENSE™

Use test lead, Part No. 3824811, for the engine-located 3-pin Deutsch connector. Disconnect the ECM A and B connectors.

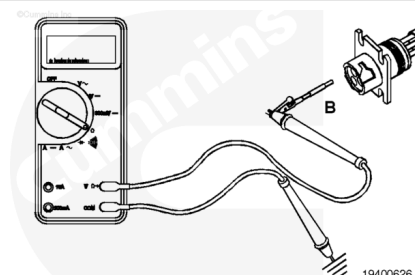
Measure the resistance from pin A of the 3-pin Deutsch connector to the engine block ground. The multimeter **must** show an open circuit (10 ohms or less).

If the circuit is **not** open, repair or replace the CENSE™ harness.
Refer to Procedure 019-043
(/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).



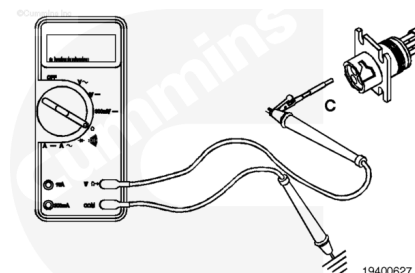
Measure the resistance from pin B of the 3-pin Deutsch connector to the engine block ground. The multimeter **must** show an open circuit (100k ohms or more).

If the circuit is **not** open, repair or replace the CENSE™ harness.
Refer to Procedure 019-043
(/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).



Measure the resistance from pin C of the 3-pin Deutsch connector to the engine block ground. The multimeter **must** show an open circuit (100k ohms or more).

If the circuit is **not** open, repair or replace the CENSE™ harness.
Refer to Procedure 019-043.
(/qs3/pubsys2/xml/en/procedures/81/81-019-043.html)

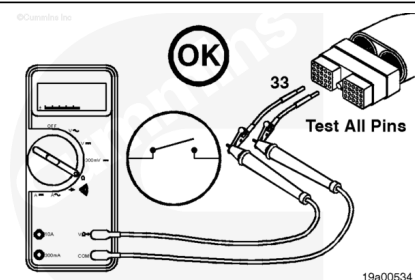


Check for Short Circuit from Pin to Pin

Engine Harness Connector

Disconnect the CENSE™ 40-pin A and B Deutsch connectors from the ECM. Disconnect the 31-pin and 23-pin OEM connectors from the OEM harness.

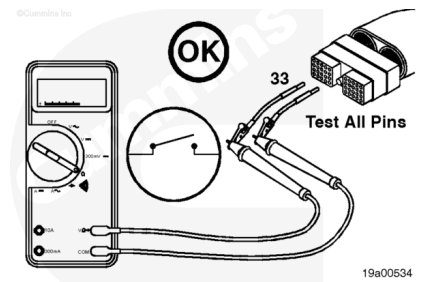
Use test lead, Part No. 3822758, for the CENSE™ ECM 40-pin Deutsch connectors. Measure the resistance from pin 33 of the ECM B connector to all other pins in the ECM B connector. The multimeter **must** show an open circuit (100k ohms or more).



If the circuit is **not** open, repair or replace the CENSE™ harness.
Refer to Procedure 019-043
(/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).

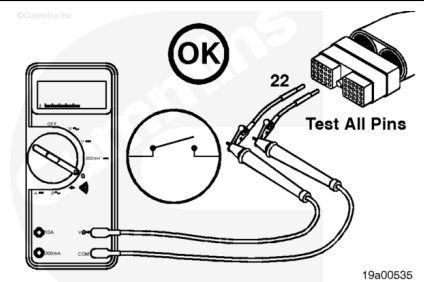
Now measure the resistance from pin 33 of the ECM B connector to all pins **in the ECM A connector**. The multimeter **must** show an open circuit (100k ohms or more).

If the circuit is **not** open, repair or replace the CENSE™ harness.
Refer to Procedure 019-043
(/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).



Measure the resistance from pin 22 of the ECM A connector to all other pins in the ECM A connector. The multimeter **must** show an open circuit (100k ohms or more).

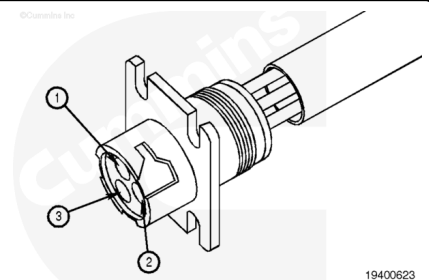
If the circuit is **not** open, repair or replace the CENSE™ harness.
Refer to Procedure 019-043
(/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).



Voltage Check

Locate the datalink connector on the CENSE™ harness.

The datalink connector is shown.

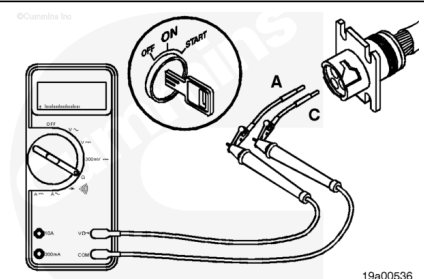


⚠ CAUTION ⚠

To avoid pin and harness damage, use test lead, Part No. 3824811.

Select the VDC function on the multimeter.

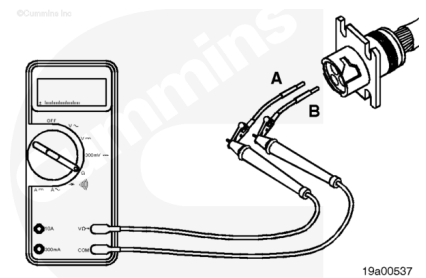
Measure the voltage from pin C of the 3-pin Deutsch connector to pin A. The multimeter **must** show -10.0 VDC (minimum -8.5 to maximum -11.0 VDC).



If no voltage is present, the datalink circuit **must** be checked for an open circuit. Verify that the battery voltage is correct.

Measure the voltage from pin B of the 3-pin Deutsch connector to pin A.

The multimeter **must** show 0 VDC. If a voltage is present, the datalink **must** be checked for a short circuit from pin to pin.



Measure the voltage from pin A of the 3-pin Deutsch connector to the engine block.

The multimeter **must** show 0 VDC.

